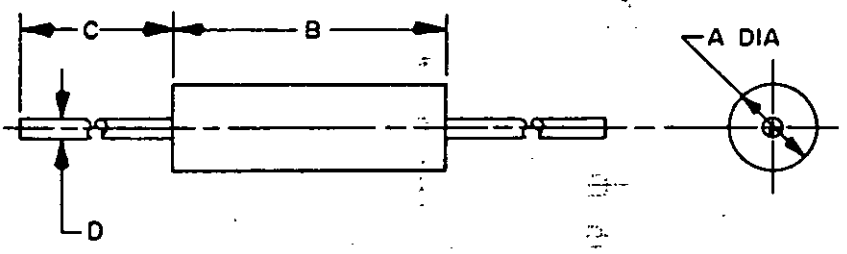
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE (IRON CORE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.085 (2.16)	.105 (2.67)
B	.240 (6.10)	.260 (6.60)
C	1.250 (31.75)	1.626 (41.30)
D	.0185 (.47)	.0215 (.55)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Iron core.

Weight: 0.30 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 test voltage: 1,000 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 300 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

DC resistance: See table I.

FSC 5950

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Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part number: M39010/09 (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A1R1**	1.1	5	25	7.9	150	.18	590
A1R2**	1.2	5,10	25	7.9	150	.18	590
A1R3**	1.3	5	25	7.9	140	.22	535
A1R5**	1.5	5,10	28	7.9	140	.22	535
A1R6**	1.6	5	28	7.9	125	.30	455
A1R8**	1.8	5,10	30	7.9	125	.30	455
A2R0**	2.0	5	30	7.9	115	.40	395
A2R2**	2.2	5,10	30	7.9	115	.40	395
A2R4**	2.4	5	30	7.9	100	.55	335
A2R7**	2.7	5,10	37	7.9	100	.55	335
A3R0**	3.0	5	37	7.9	90	.85	270
A3R3**	3.3	5,10	45	7.9	90	.85	270
A3R6**	3.6	5	45	7.9	80	1.00	250
A3R9**	3.9	5,10	45	7.9	80	1.00	250
A4R3**	4.3	5	45	7.9	75	1.20	230
A4R7**	4.7	5,10	45	7.9	75	1.20	230
A5R1**	5.1	5	45	7.9	65	1.80	185
A5R6**	5.6	5,10	50	7.9	65	1.80	185
A6R2**	6.2	5	50	7.9	60	2.00	175
A6R8**	6.8	5,10	50	7.9	60	2.00	175
A7R5**	7.5	5	50	7.9	55	2.70	155
A8R2**	8.2	5,10	55	7.9	55	2.70	155
A9R1**	9.1	5	55	7.9	50	3.70	130
A100**	10.0	5,10	55	7.9	50	3.70	130
A110**	11.0	5	45	2.5	40	2.70	130

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A120**	12.0	5,10	45	2.5	40	2.70	155
A130**	13.0	5	40	2.5	35	2.80	150
A150**	15.0	5,10	40	2.5	35	2.80	150
A160**	16.0	5	40	2.5	30	3.10	145
A180**	18.0	5,10	50	2.5	30	3.10	145
A200**	20.0	5	50	2.5	25	3.30	140
A220**	22.0	5,10	50	2.5	25	3.30	140
A240**	24.0	5	50	2.5	20	3.50	135
A270**	27.0	5,10	50	2.5	20	3.50	135

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.
2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable % from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±2	---	---	-10
Group IV	±5	±(3%+.001 ohm)	1/	-10
Group V	±5	±(2%+.001 ohm)	1/	-15
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(2%+.001 ohm)	1/	-15
Subgroup 3	±2	---	---	-10
Subgroup 4	±5	±(3%+.001 ohm)	1/	-10

1/ The self-resonant frequency shall be not less than the value specified in table I.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Army - ER

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